Audio Visualiser DSP Board

https://github.com/drewsum/audio_visualiser

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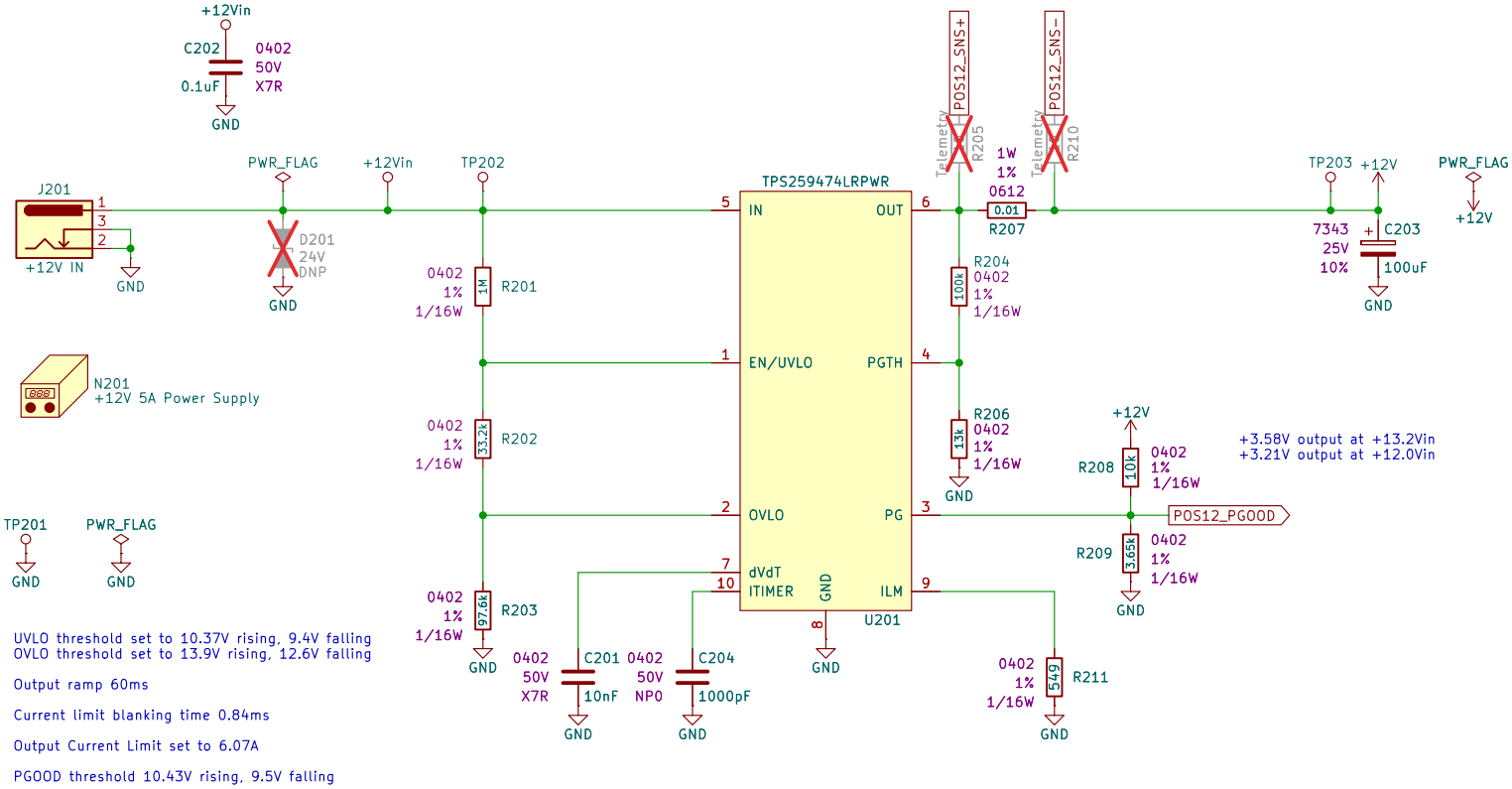
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Sheet: /
File: Audio_Visualiser_DSP.kicad_sch

Title: Audio Visualiser DSP

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Load	Draw	Quantity	Total
LED Panel VLED 256 * 0.02A 8 3V = 15.36W / 12V = 1.28A / 80% efficiency	1.6A	2	3.2A
LED Panel POS3P3 1A * 3.3V / 12V / 80% efficiency	0.344A	2	0.688A
Local POS3P3 1A * 3.3V / 12V / 80% efficiency	0.344A	1	0.344A
Total		50.784W	4.232A

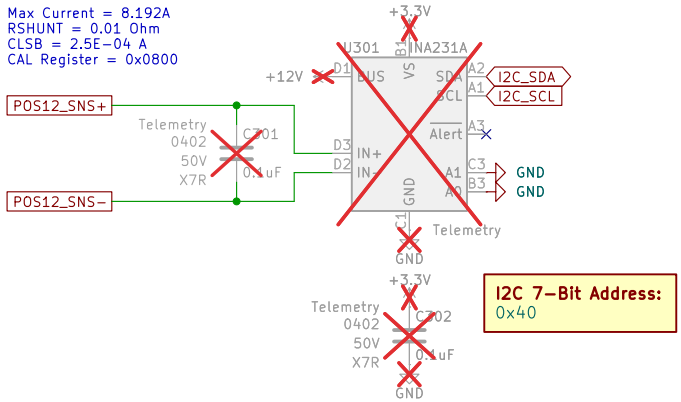
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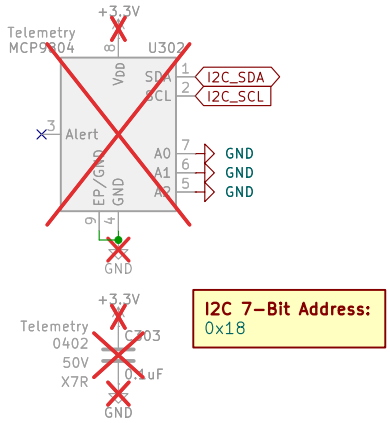
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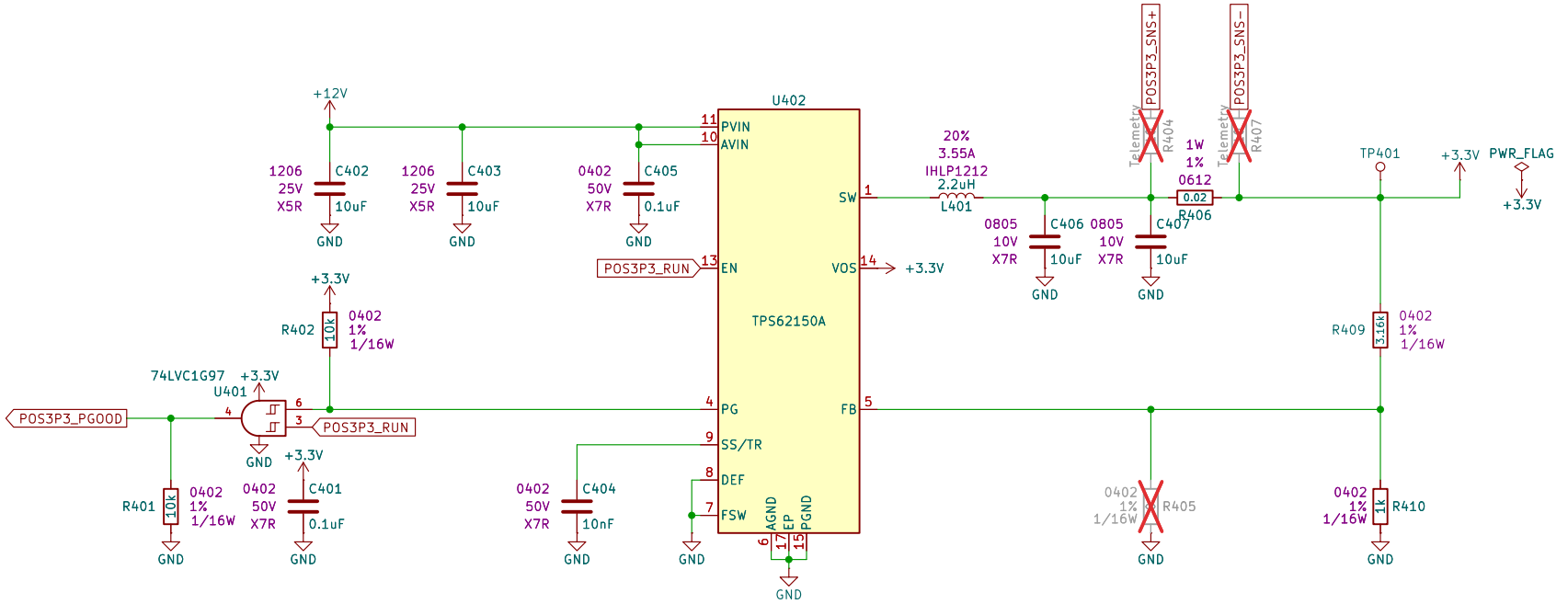
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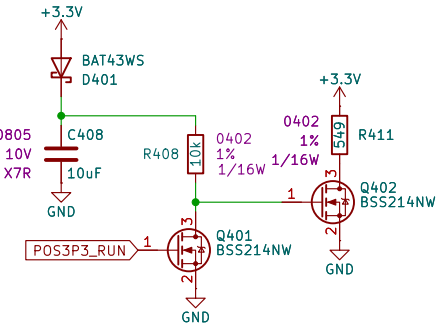




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Title: Audio Visualiser DSP		
Size: A	Date: 2026-10-27	**Rev: A**
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Load	Draw	Quantity	Total
dsPIC	50mA	1	50mA
8MHz dsPIC Clock	7mA	1	7mA
MCP6N11	1.1mA	6	6.6mA
MCP6024	5.4mA	3	16.2mA
Analog Switch	0.05mA	1	0.05mA
Digipot	0.6mA	1	0.6mA
1.6V Reference	0.2mA	1	0.2mA
PCA9536 GPIO Expander	0.29mA	1	0.29mA
I2C Buffers	8.3mA	3	24.9mA
Octal Buffers	1mA	2	2mA
LVC1G19s	0.01mA	21	0.21mA
Rotary Encoder Module	50mA	1	50mA
LEDs	1.3mA	11	14.3mA
INA231	0.42mA	2	0.84mA
MCP9804	0.4mA	3	1.2mA
ETC	0.36mA	1	0.36mA
Cap Touch Sensor	0.729mA	1	0.729mA
Total		0.579W	175.48mA

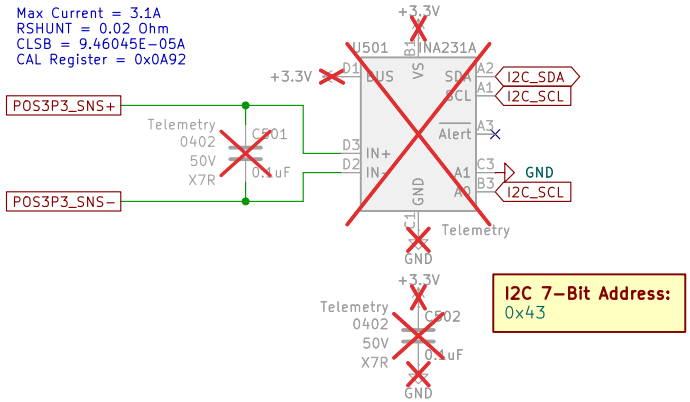
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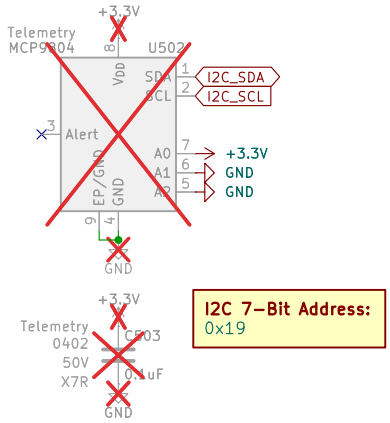
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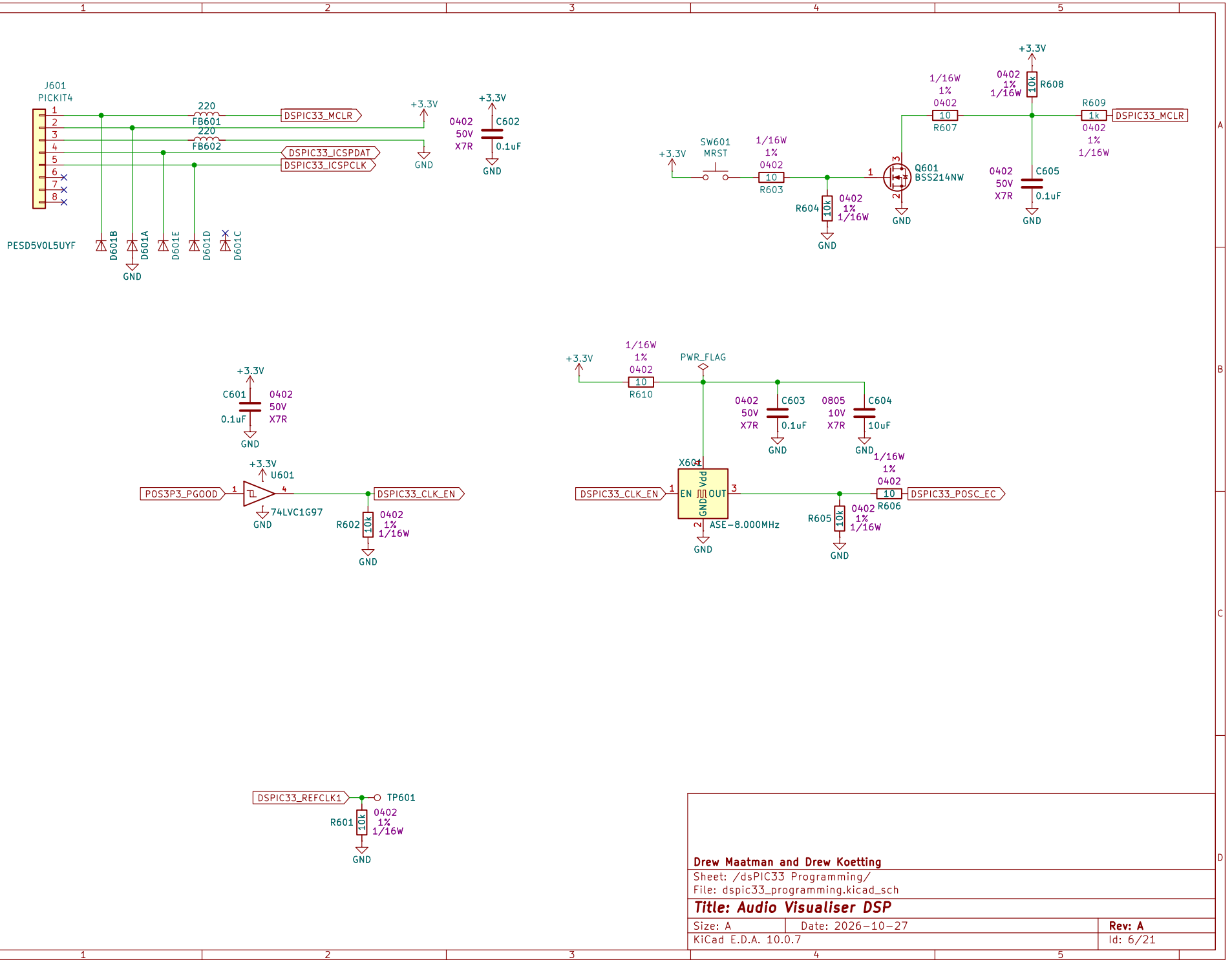
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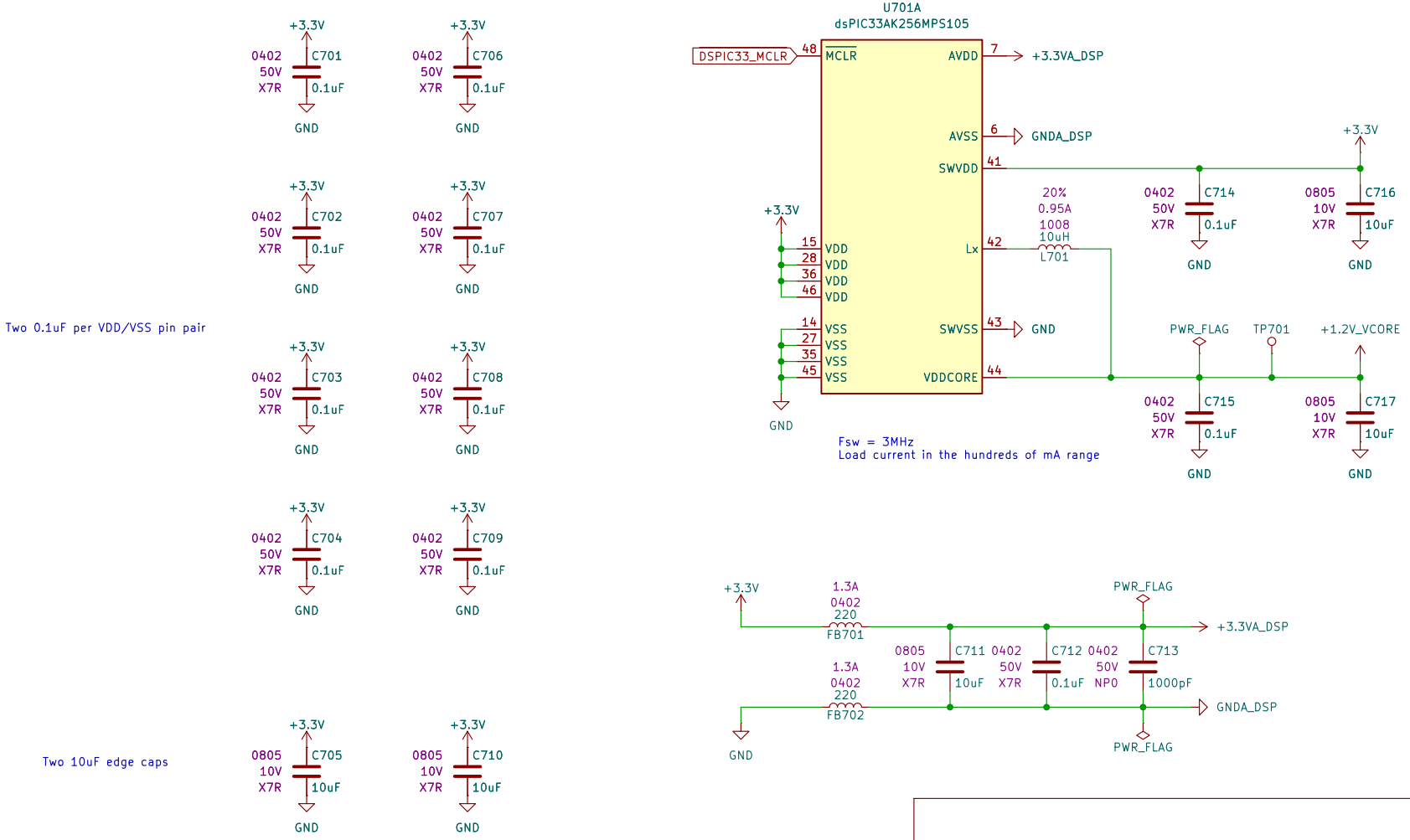
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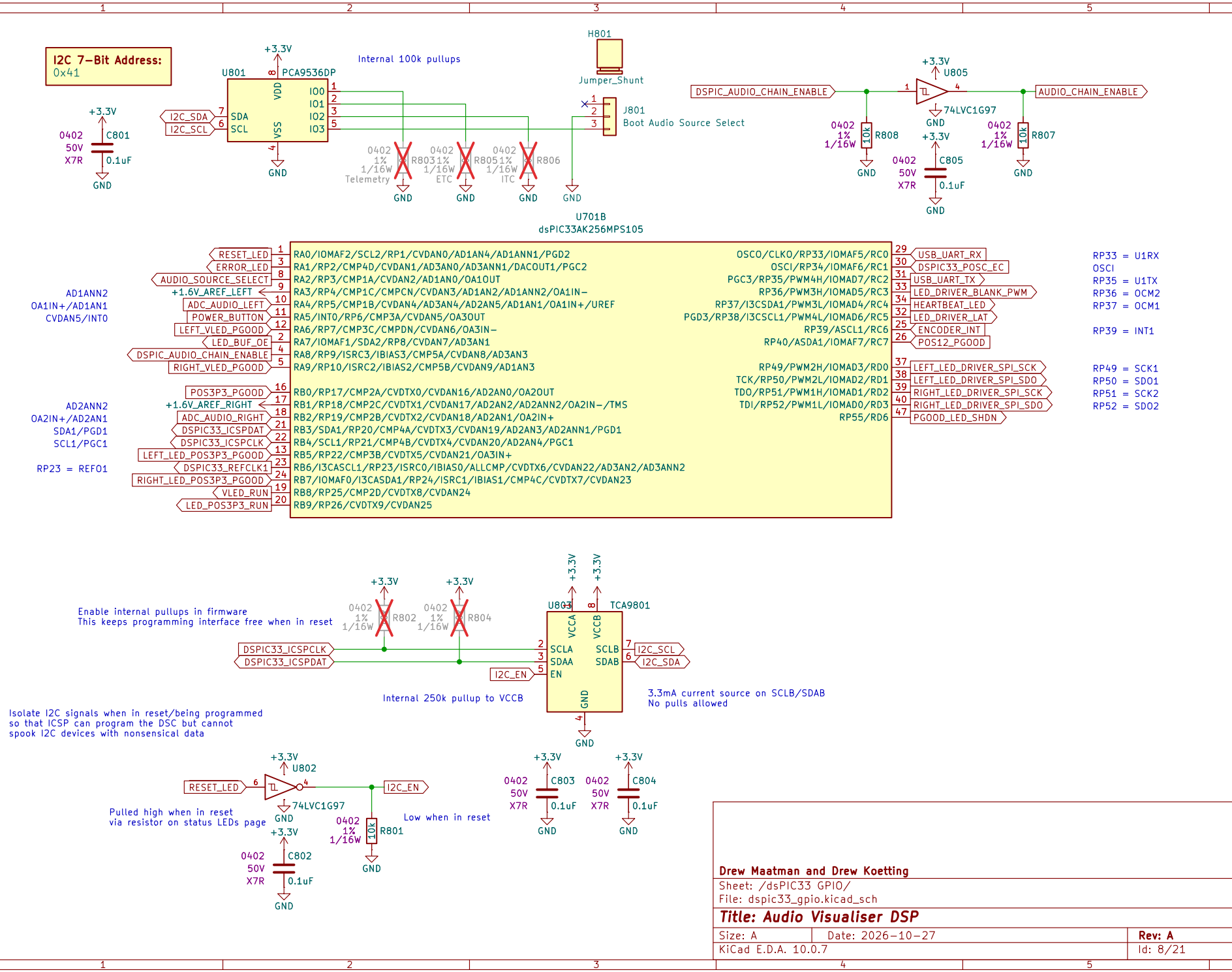


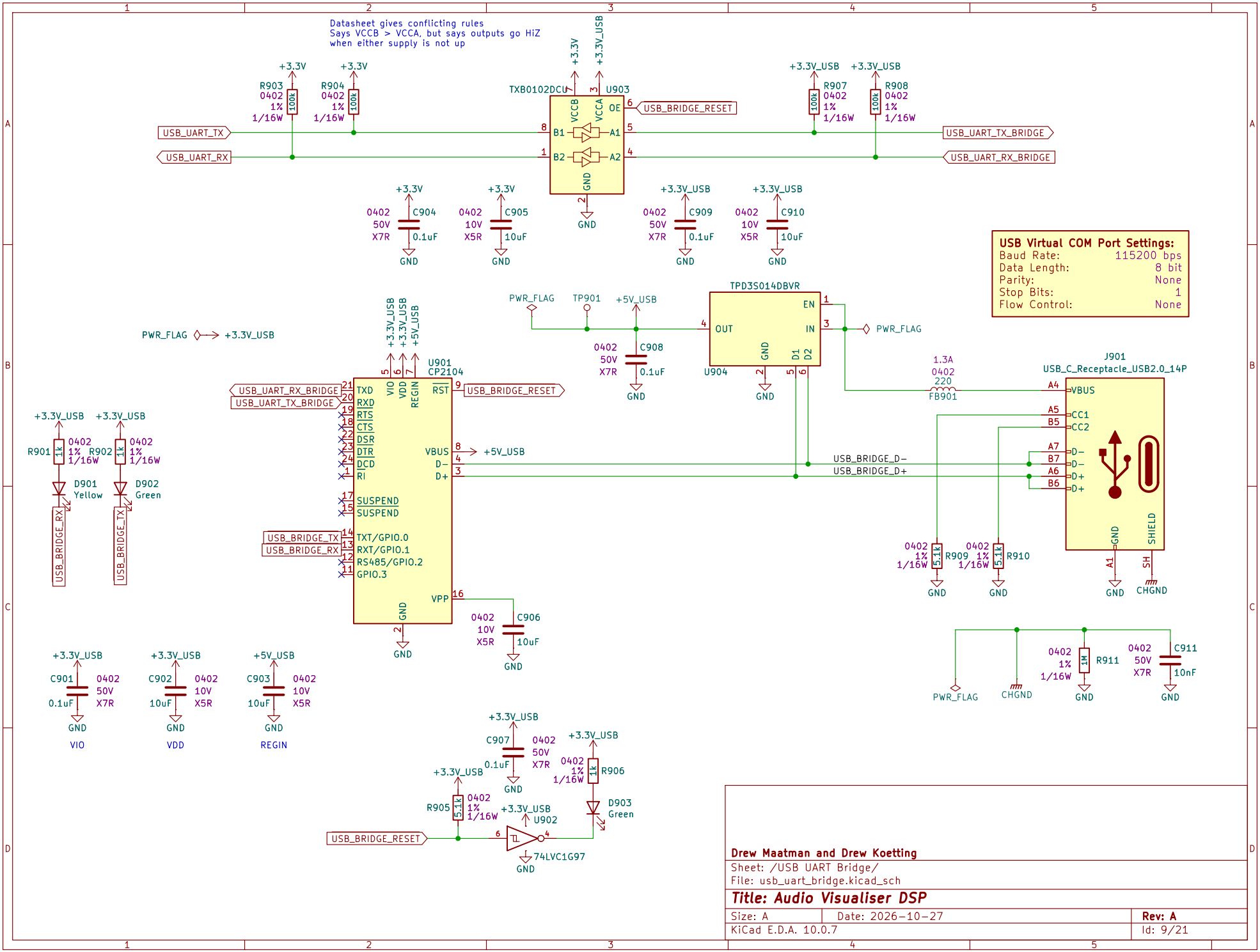
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Title: Audio Visualiser DSP		
Size: A	Date: 2026-10-27	**Rev: A**
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J601
PICKIT4
1
2
3
4
5
6
7
8
220
FB601
220
FB602
DSPIC33_MCLR
DSPIC33_ICSPDAT
DSPIC33_ICSPCLK
+3.3V
GND
+3.3V
0402
50V
X7R
C602
0.1uF
GND
PESD5V0L5UYF
D601B
D601A
D601E
D601D
D601C
GND
+3.3V
1/16W
1%
0402
10
R607
0402
1%
1/16W
10k
R608
R609
1k
DSPIC33_MCLR
0402
1%
1/16W
+3.3V
SW601
MRST
1/16W
1%
0402
10
R603
1
Q601
BSS214NW
2
3
R604
10k
0402
1%
1/16W
GND
GND
0402
50V
X7R
C605
0.1uF
GND
+3.3V
C601
0402
50V
X7R
0.1uF
GND
+3.3V
U601
POS3P3_PGOOD
1
TL
4
74LVC1G97
GND
DSPIC33_CLK_EN
R602
10k
0402
1%
1/16W
GND
+3.3V
1/16W
1%
0402
10
R610
PWR_FLAG
0402
50V
X7R
C603
0.1uF
GND
0805
10V
X7R
C604
10uF
GND
X601
DSPIC33_CLK_EN
1
EN
Vdd
OUT
3
GND
2
ASE-8.000MHz
GND
1/16W
1%
0402
10
R606
DSPIC33_POSC_EC
R605
10k
0402
1%
1/16W
GND
DSPIC33_REFCLK1
TP601
R601
10k
0402
1%
1/16W
GND
Drew Maatman and Drew Koetting
Sheet: /dsPIC33 Programming/
File: dspic33_programming.kicad_sch
Title: Audio Visualiser DSP
Size: A
Date: 2026-10-27
Rev: A
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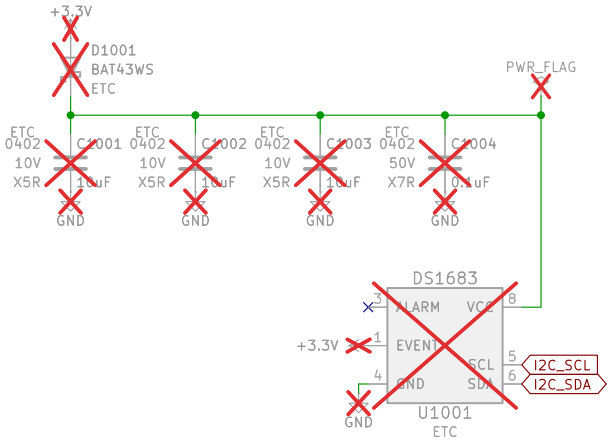


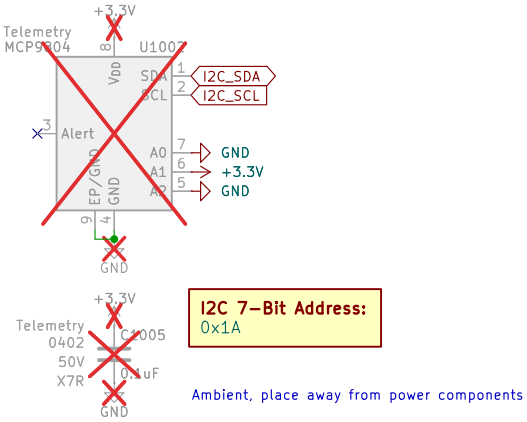
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Title: Audio Visualiser DSP		
Size: A	Date: 2026-10-27	**Rev: A**
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I2C 7-Bit Address:
0x41
U801
PCA9536DP
Internal 100k pullups
H801
Jumper_Shunt
J801
Boot Audio Source Select
U805
74LVC1G97
DSPIC_AUDIO_CHAIN_ENABLE
AUDIO_CHAIN_ENABLE
R808
R807
C805
C801
R8031%
R8051%
R806
Telemetry
ETC
ITC
U701B
dsPIC33AK256MPS105
RESET_LED
ERROR_LED
AUDIO_SOURCE_SELECT
+1.6V_AREF_LEFT
ADC_AUDIO_LEFT
POWER_BUTTON
LEFT_VLED_PGOOD
LED_BUF_OE
DSPIC_AUDIO_CHAIN_ENABLE
RIGHT_VLED_PGOOD
POS3P3_PGOOD
+1.6V_AREF_RIGHT
ADC_AUDIO_RIGHT
DSPIC33_ICSPDAT
DSPIC33_ICSPCLK
LEFT_LED_POS3P3_PGOOD
DSPIC33_REFCLK1
RIGHT_LED_POS3P3_PGOOD
VLED_RUN
LED_POS3P3_RUN
RA0/IOMAF2/SCL2/RP1/CVDAN0/AD1AN4/AD1ANN1/PGD2
RA1/RP2/CMP4D/CVDAN1/AD3AN0/AD3ANN1/DACOUT1/PGC2
RA2/RP3/CMP1A/CVDAN2/AD1AN0/OA1OUT
RA3/RP4/CMP1C/CMPCN/CVDAN3/AD1AN2/AD1ANN2/OA1IN−
RA4/RP5/CMP1B/CVDAN4/AD3AN4/AD2AN5/AD1AN1/OA1IN+/UREF
RA5/INT0/RP6/CMP3A/CVDAN5/OA3OUT
RA6/RP7/CMP3C/CMPDN/CVDAN6/OA3IN−
RA7/IOMAF1/SDA2/RP8/CVDAN7/AD3AN1
RA8/RP9/ISRC3/IBIAS3/CMP5A/CVDAN8/AD3AN3
RA9/RP10/ISRC2/IBIAS2/CMP5B/CVDAN9/AD1AN3
RB0/RP17/CMP2A/CVDTX0/CVDAN16/AD2AN0/OA2OUT
RB1/RP18/CMP2C/CVDTX1/CVDAN17/AD2AN2/AD2ANN2/OA2IN−/TMS
RB2/RP19/CMP2B/CVDTX2/CVDAN18/AD2AN1/OA2IN+
RB3/SDA1/RP20/CMP4A/CVDTX3/CVDAN19/AD2AN3/AD2ANN1/PGD1
RB4/SCL1/RP21/CMP4B/CVDTX4/CVDAN20/AD2AN4/PGC1
RB5/RP22/CMP3B/CVDTX5/CVDAN21/OA3IN+
RB6/I3CASCL1/RP23/ISRC0/IBIAS0/ALLCMP/CVDTX6/CVDAN22/AD3AN2/AD3ANN2
RB7/IOMAF0/I3CASDA1/RP24/ISRC1/IBIAS1/CMP4C/CVDTX7/CVDAN23
RB8/RP25/CMP2D/CVDTX8/CVDAN24
RB9/RP26/CVDTX9/CVDAN25
OSCO/CLKO/RP33/IOMAF5/RC0
OSCI/RP34/IOMAF6/RC1
PGC3/RP35/PWM4H/IOMAD7/RC2
RP36/PWM3H/IOMAD5/RC3
RP37/I3CSDA1/PWM3L/IOMAD4/RC4
PGD3/RP38/I3CSCL1/PWM4L/IOMAD6/RC5
RP39/ASCL1/RC6
RP40/ASDA1/IOMAF7/RC7
RP49/PWM2H/IOMAD3/RD0
TCK/RP50/PWM2L/IOMAD2/RD1
TDO/RP51/PWM1H/IOMAD1/RD2
TDI/RP52/PWM1L/IOMAD0/RD3
RP55/RD6
USB_UART_RX
DSPIC33_POSC_EC
USB_UART_TX
LED_DRIVER_BLANK_PWM
HEARTBEAT_LED
LED_DRIVER_LAT
ENCODER_INT
POS12_PGOOD
LEFT_LED_DRIVER_SPI_SCK
LEFT_LED_DRIVER_SPI_SDO
RIGHT_LED_DRIVER_SPI_SCK
RIGHT_LED_DRIVER_SPI_SDO
PGOOD_LED_SHDN
RP33 = U1RX
OSCI
RP35 = U1TX
RP36 = OCM2
RP37 = OCM1
RP39 = INT1
RP49 = SCK1
RP50 = SDO1
RP51 = SCK2
RP52 = SDO2
AD1ANN2
OA1IN+/AD1AN1
CVDAN5/INT0
AD2ANN2
OA2IN+/AD2AN1
SDA1/PGD1
SCL1/PGC1
RP23 = REFO1
Enable internal pullups in firmware
This keeps programming interface free when in reset
R802
R804
U803
TCA9801
I2C_EN
Internal 250k pullup to VCCB
3.3mA current source on SCLB/SDAB
No pulls allowed
Isolate I2C signals when in reset/being programmed
so that ICSP can program the DSC but cannot
spook I2C devices with nonsensical data
U802
Pulled high when in reset
via resistor on status LEDs page
Low when in reset
R801
C802
C803
C804
Drew Maatman and Drew Koetting
Sheet: /dsPIC33 GPIO/
File: dspic33_gpio.kicad_sch
Title: Audio Visualiser DSP
Size: A
Date: 2026-10-27
Rev: A
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Id: 8/21



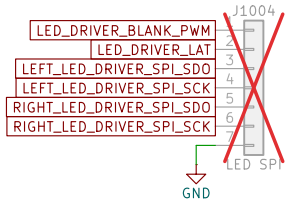
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Size: A	Date: 2026-10-27	**Rev: A**
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Test Points



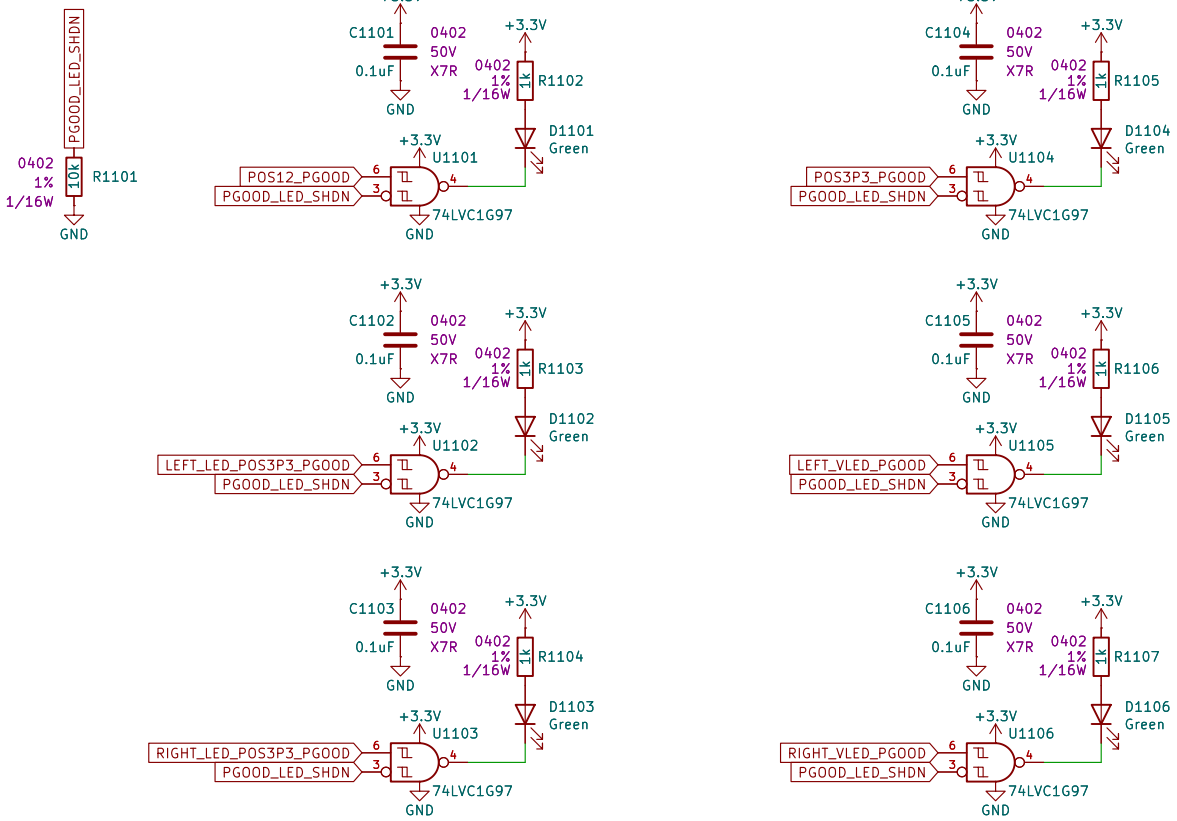


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File: platform_etc.kicad_sch

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KiCad E.D.A. 10.0.7		Id: 10/21

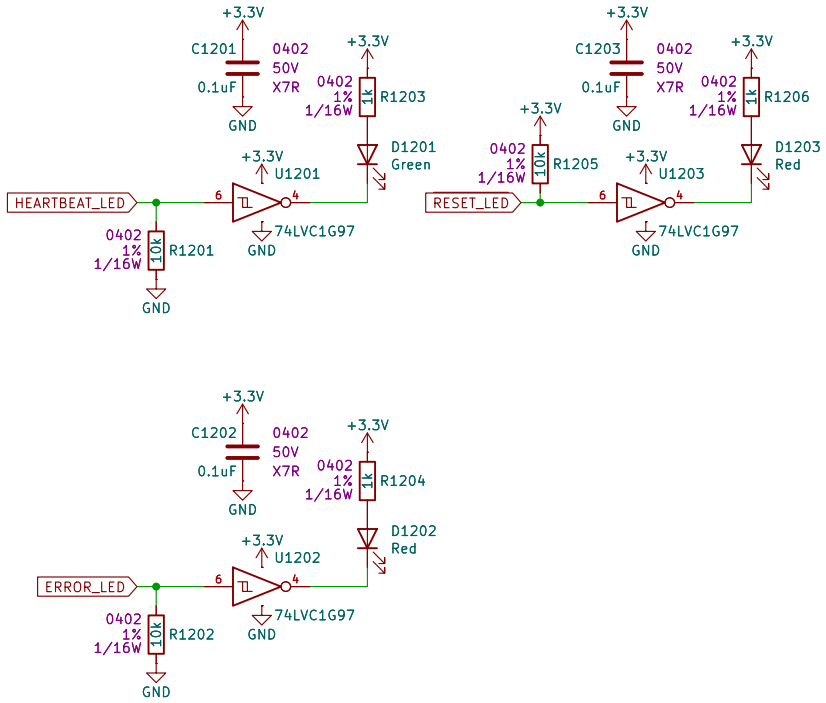


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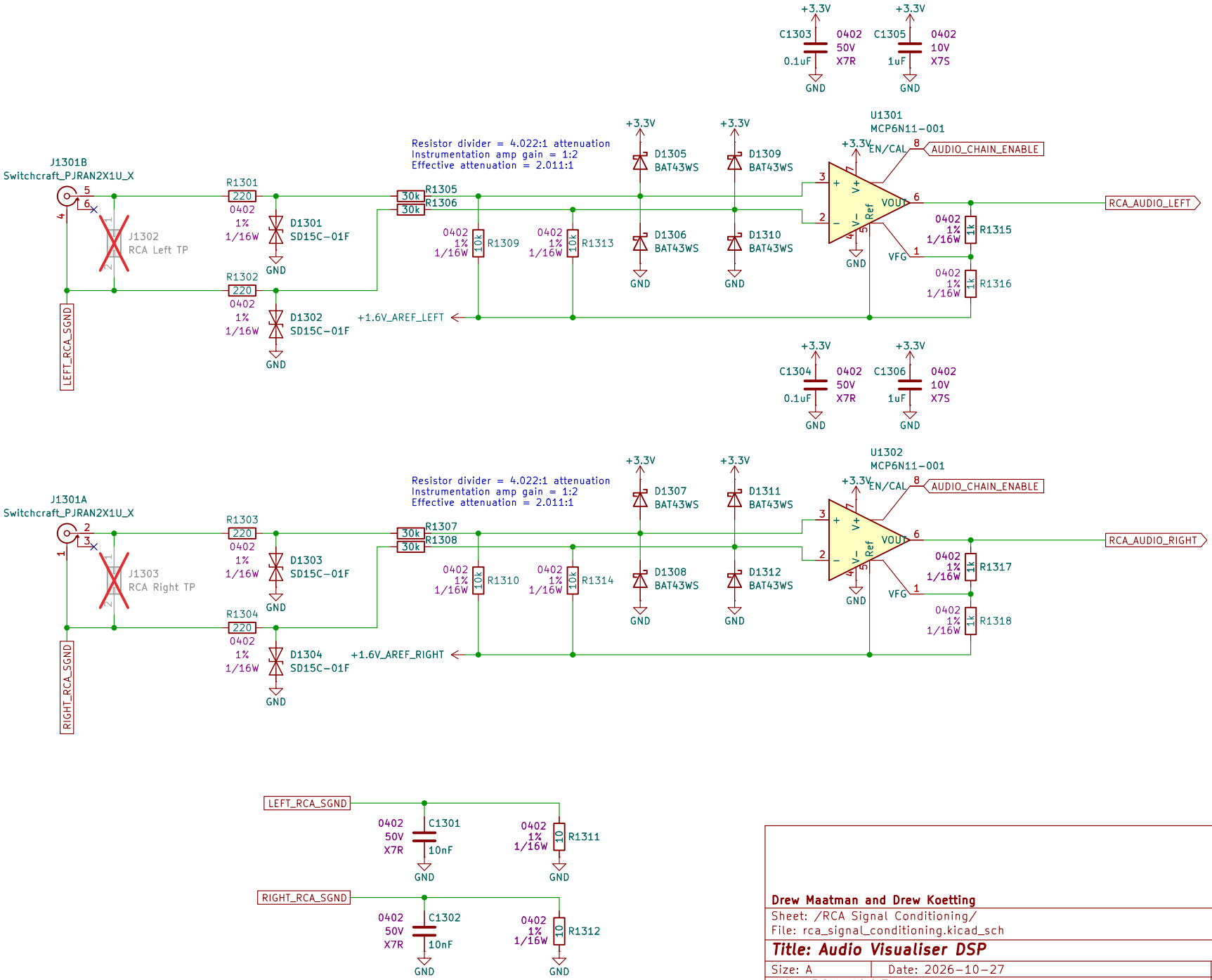
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Title: Audio Visualiser DSP

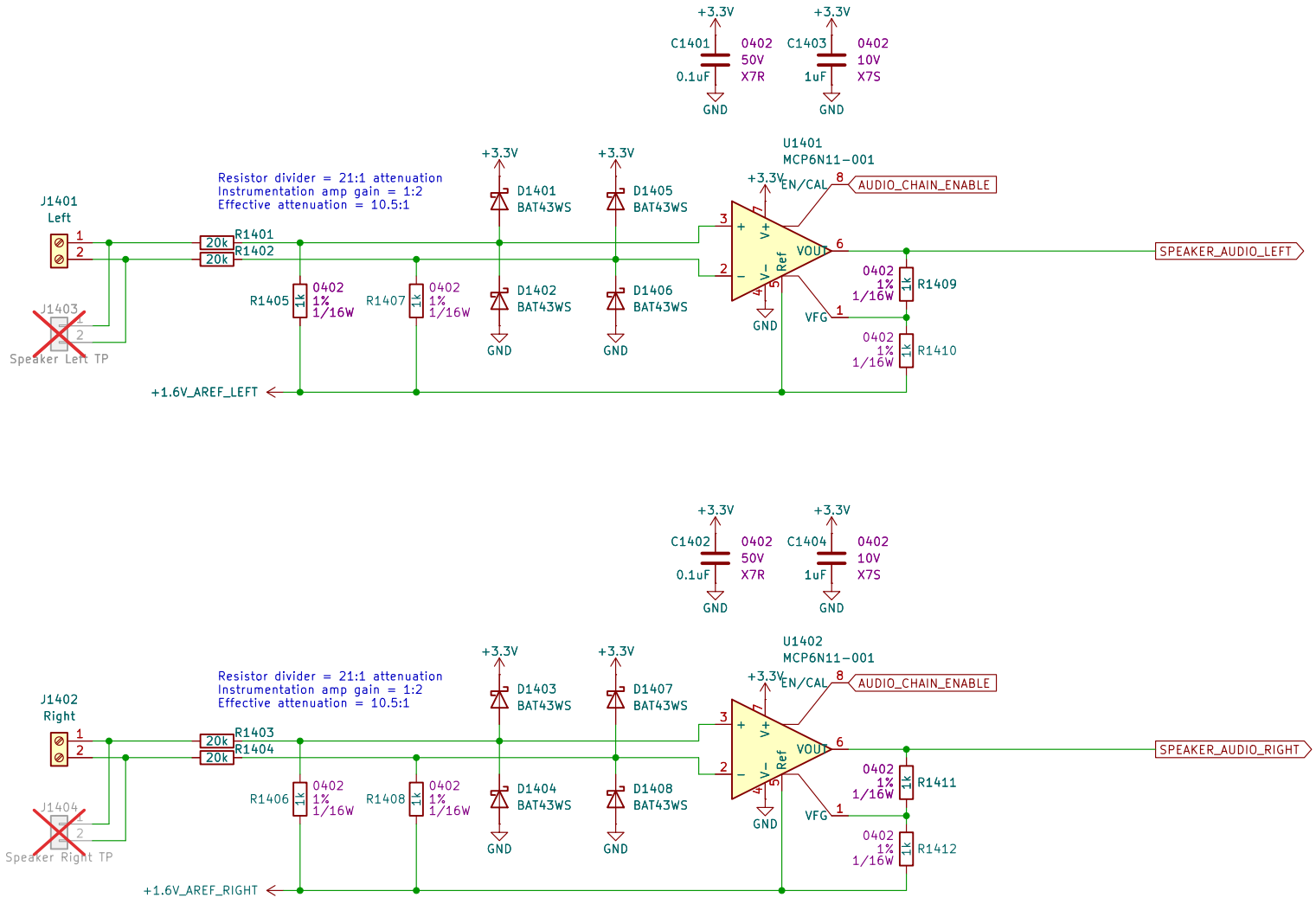
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KiCad E.D.A. 10.0.7		Id: 11/21



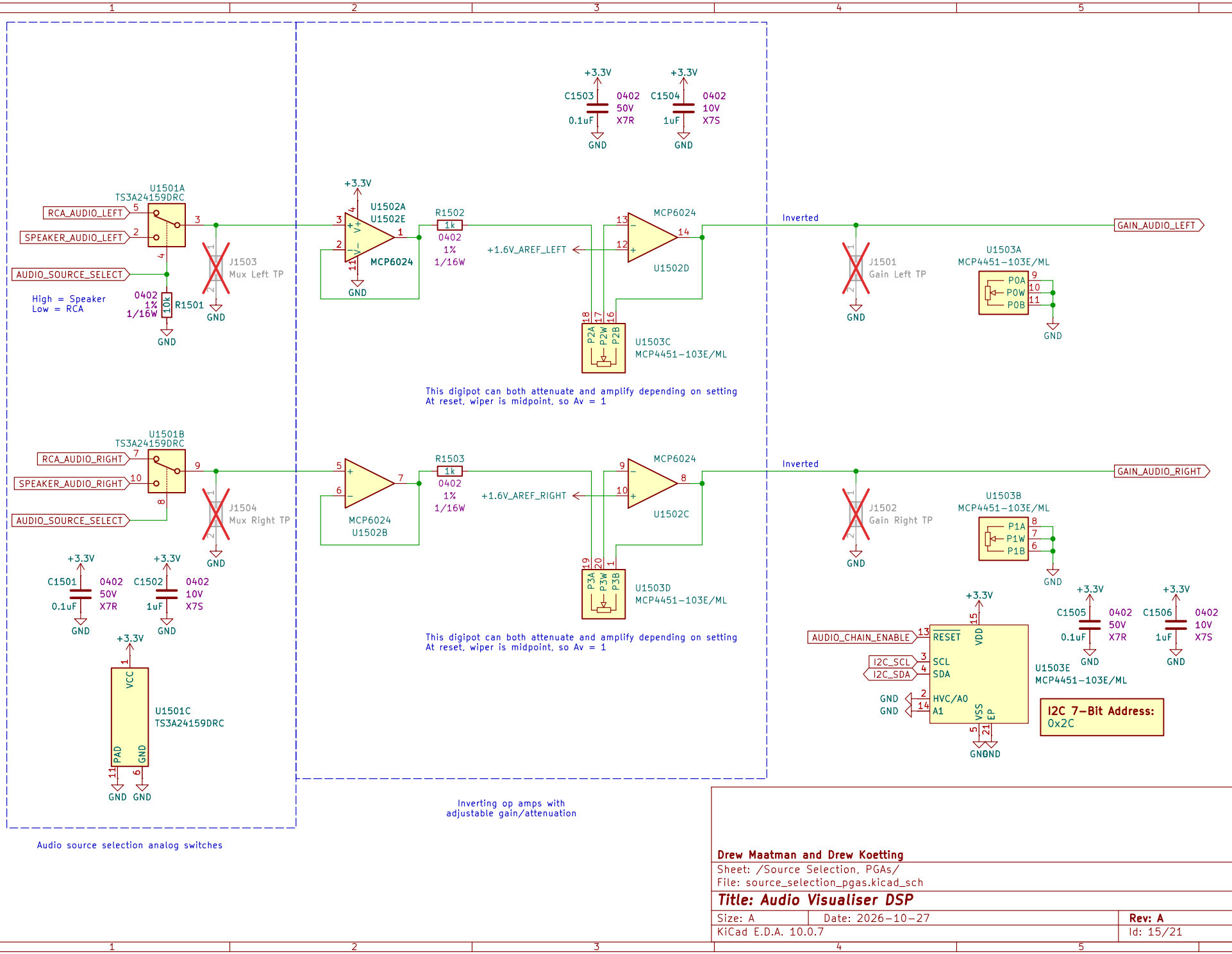
Drew Maatman and Drew Koetting		
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Title: Audio Visualiser DSP		
Size: A	Date: 2026-10-27	**Rev: A**
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Drew Maatman and Drew Koetting		
Sheet: /RCA Signal Conditioning/		
File: rca_signal_conditioning.kicad_sch		
Title: Audio Visualiser DSP		
Size: A	Date: 2026-10-27	**Rev: A**
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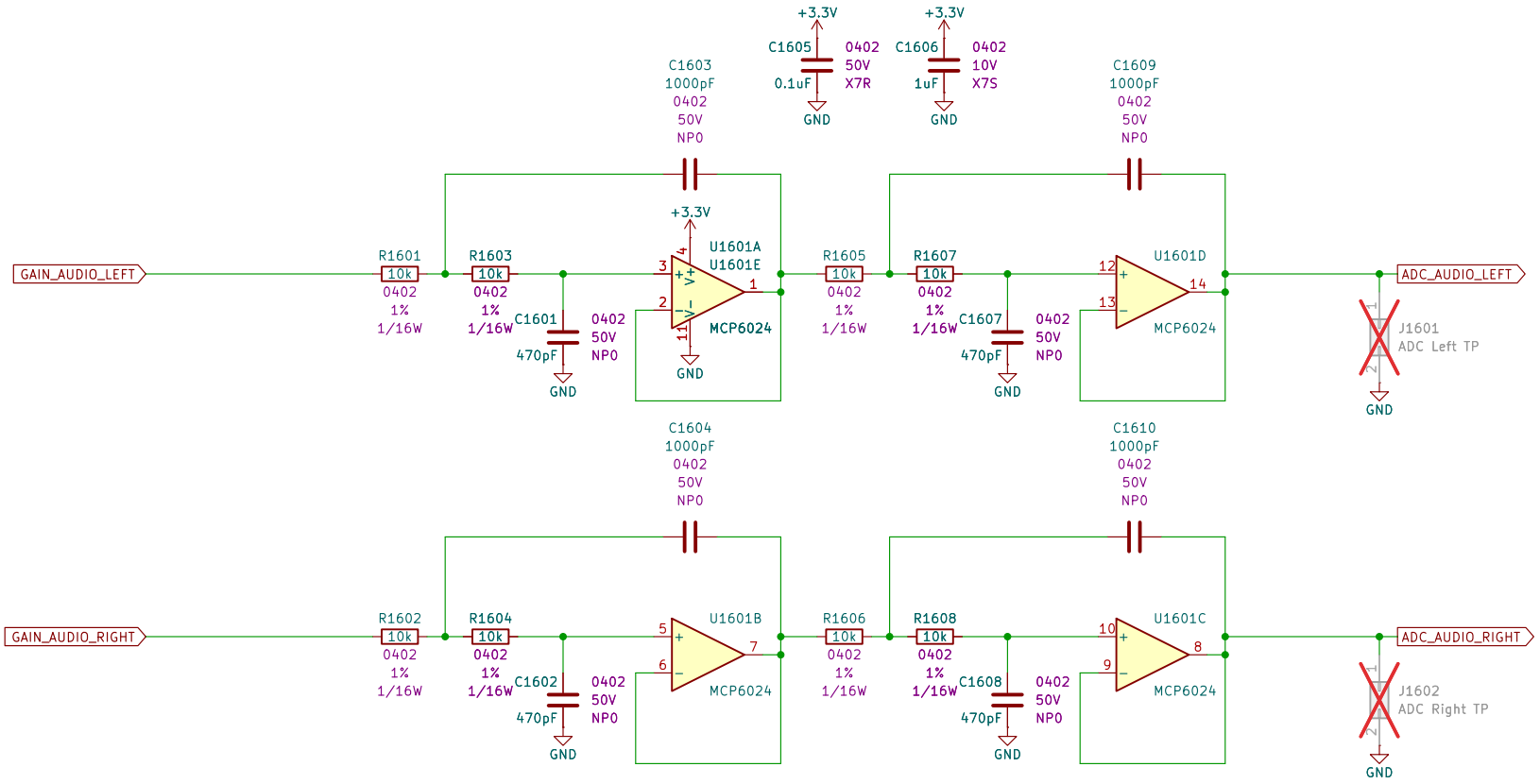


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Size: A	Date: 2026-10-27	**Rev: A**
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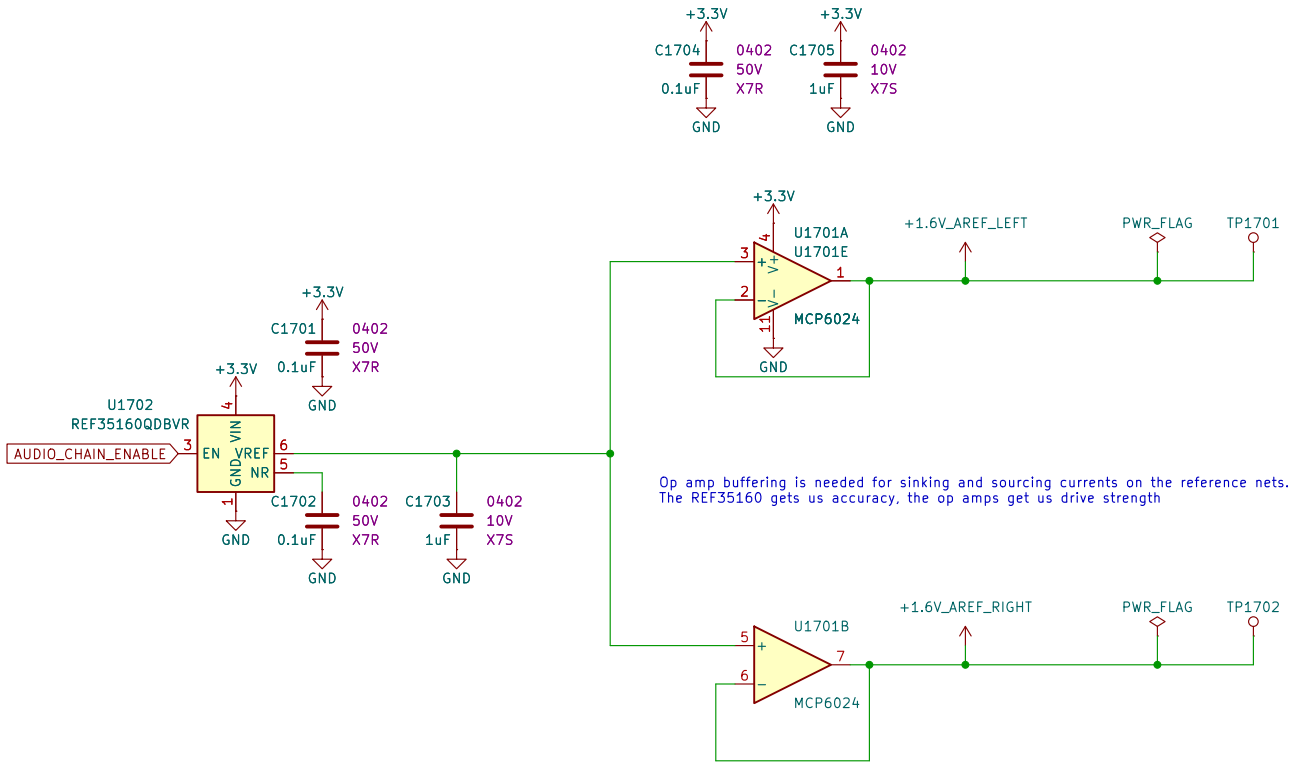
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Sheet: /Source Selection, PGAs/	
File: source_selection_pgas.kicad_sch	
Title: Audio Visualiser DSP	
Size: A	Date: 2026-10-27
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Rev: A



4-pole Sallen-Key active anti-aliasing lowpass filters
Unity gain
fc = 23.2kHz, Q = 0.707 each
If sampling at 44.1kHz, nyquist is 22.05kHz
Must filter hard around this frequency

Drew Maatman and Drew Koetting		
Sheet: /Anti-alias Filtering/		
File: anti_alias_filtering.kicad_sch		
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Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 16/21

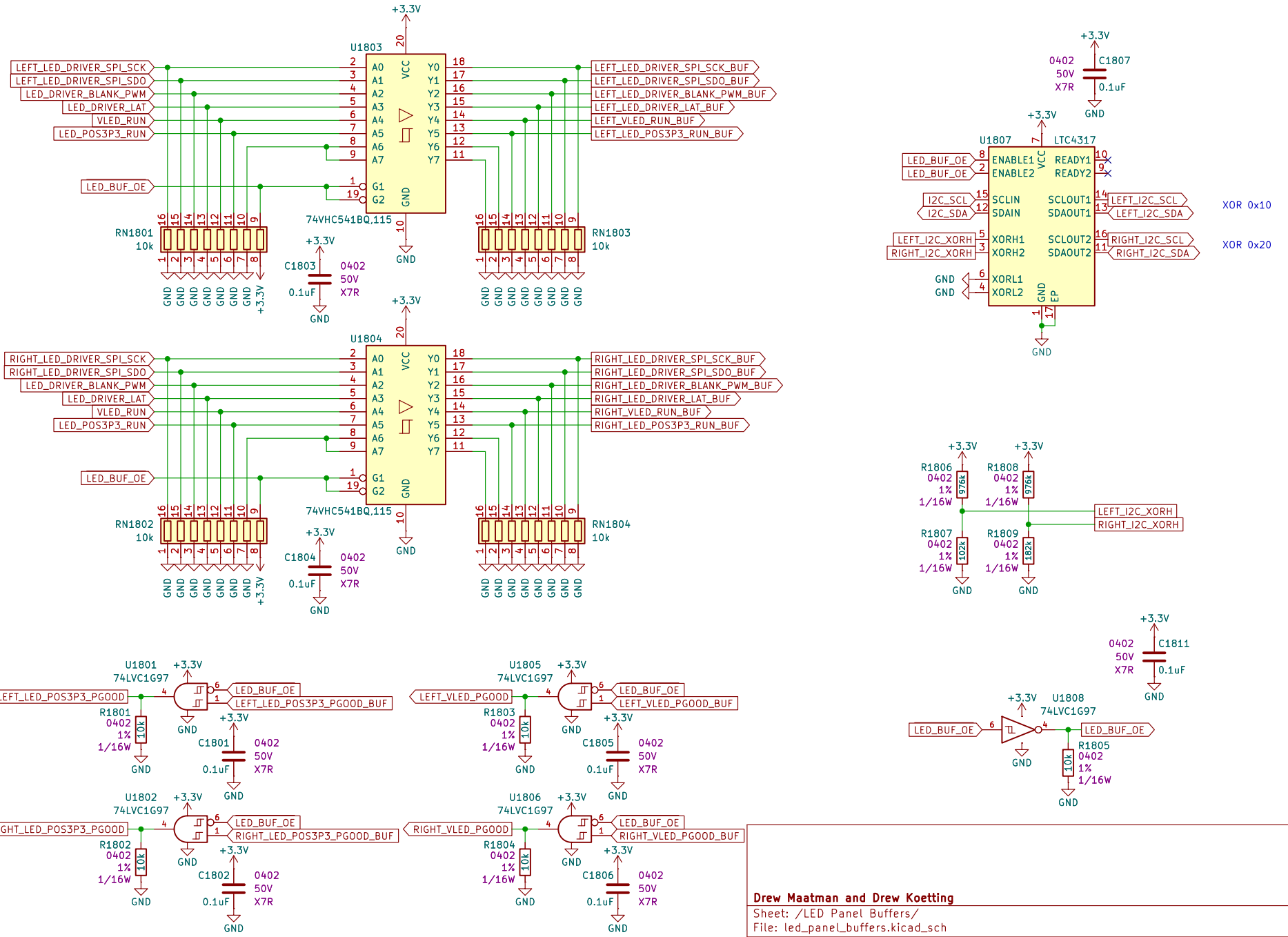


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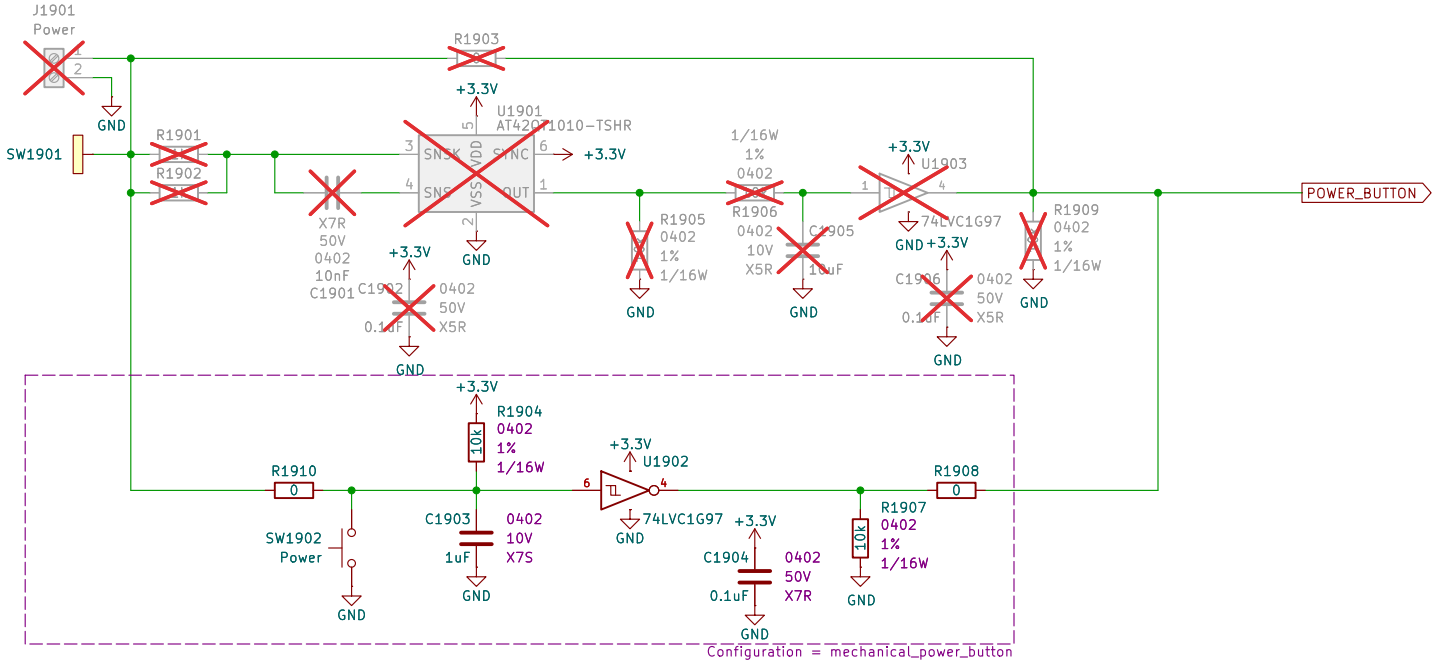
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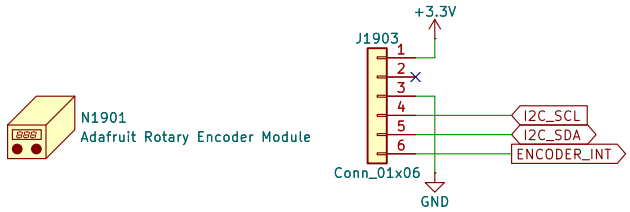
Title: Audio Visualiser DSP

Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 17/21



Drew Maatman and Drew Koetting	
Sheet: /LED Panel Buffers/	
File: led_panel_buffers.kicad_sch	
Title: Audio Visualiser DSP	
Size: A	Date: 2026-10-27
KiCad E.D.A. 10.0.7	Id: 18/21





I2C 7-Bit Address:
0x36

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Sheet: /Pushbuttons/

File: pushbuttons.kicad_sch

Title: Audio Visualiser DSP

Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 19/21

H2001
MountingHole

H2003
MountingHole

H2002
MountingHole

H2004
MountingHole

Screw
MK2001

Screw
MK2005

Screw
MK2002

Screw
MK2006

Standoff
MK2003

Standoff
MK2007

Standoff
MK2004

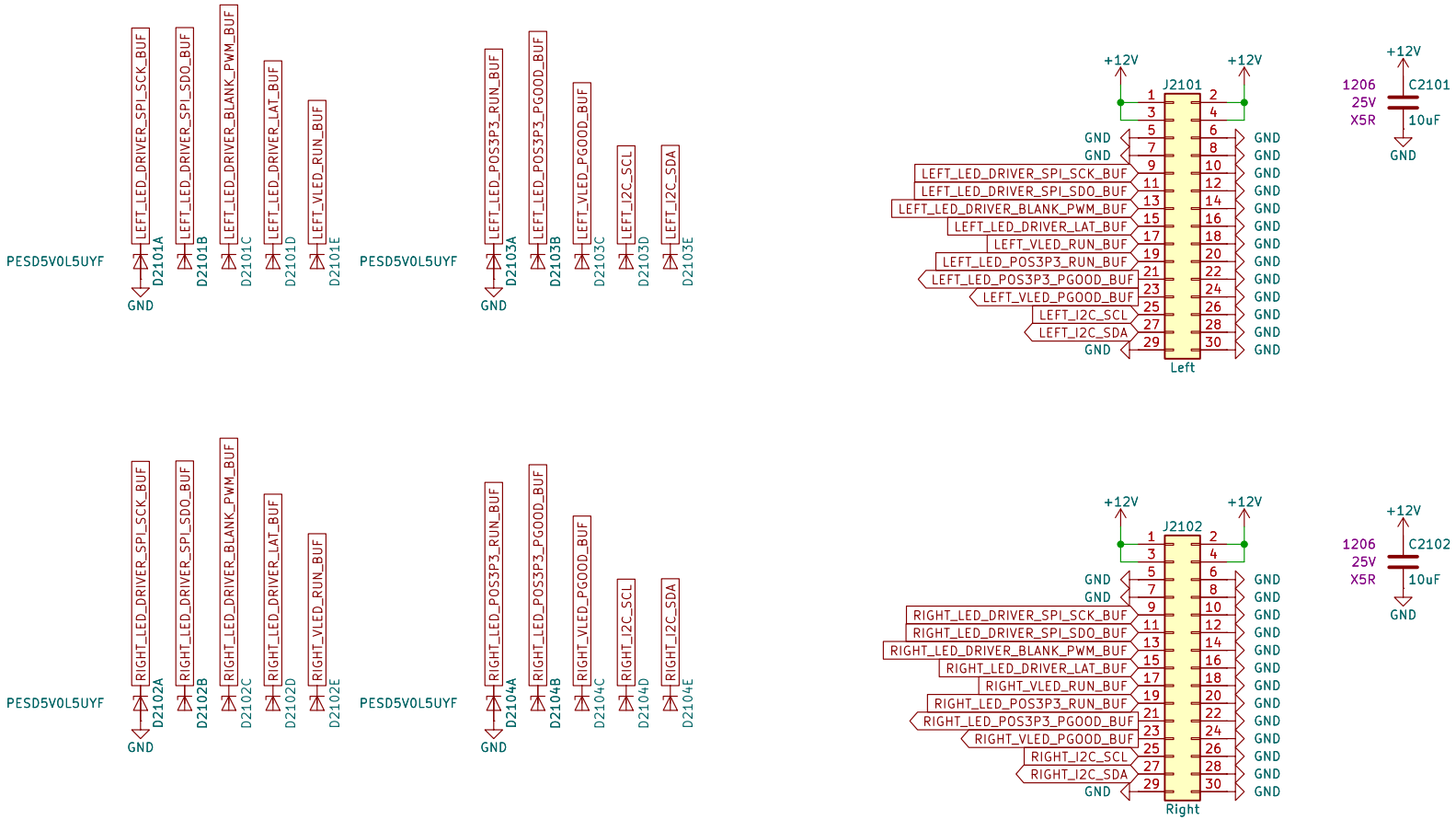
Standoff
MK2008

Drew Maatman and Drew Koetting

Sheet: /Mechanical/
File: mechanical.kicad_sch

Title: Audio Visualiser DSP

Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 20/21



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Sheet: /LED Panel Connectors/
File: led_panel_connectors.kicad_sch

Title: Audio Visualiser DSP

Size: A	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 21/21